MODULE 3.2.P — DRUG PRODUCT

AMLOZA (Amlodipine Besylate Tablets, USP) 5 mg / 10 mg — ANDA

3.2.P.1 Description and Composition

AMLOZA Tablets are white, round, biconvex, uncoated, immediate-release tablets for oral administration, available in 5 mg and 10 mg strengths (expressed as amlodipine).

Component	Function
Amlodipine Besylate USP	Active ingredient
Microcrystalline Cellulose NF	Diluent / filler
Anhydrous Dibasic Calcium Phosphate USP	Diluent / binder
Sodium Starch Glycolate, Type A NF	Disintegrant
Magnesium Stearate NF	Lubricant

The qualitative composition is designed to be consistent with the Reference Listed Drug (Norvasc Tablets); Q1 same and Q2 similar relative to the RLD would be confirmed during formal formulation development and RLD reverse-engineering studies in a real submission.

3.2.P.2 Pharmaceutical Development

3.2.P.2.1 Components of the Drug Product

3.2.P.2.1.1 Drug Substance: Based on published literature and prior use of these excipients in approved amlodipine products, no known incompatibilities have been identified between amlodipine besylate and microcrystalline cellulose, dibasic calcium phosphate, sodium starch glycolate, or magnesium stearate; however, no experimental drug–excipient compatibility data are presented in this academic dossier.

3.2.P.2.1.2 Excipients: Each excipient was selected based on established compendial status, compatibility, and prior use in approved amlodipine besylate products (as referenced in the FDA Inactive Ingredient Database).

3.2.P.2.2 Drug Product

3.2.P.2.2.1 Formulation Development: Literature reports amlodipine besylate as exhibiting high solubility and high permeability, consistent with a BCS Class I profile, with acceptable flow and compression characteristics supporting selection of direct compression over wet granulation; this ANDA is supported by in vivo bioequivalence data (Module 5) rather than a BCS-based biowaiver. Direct compression was preferred to minimize processing steps and reduce exposure of the drug substance to moisture and heat during manufacture, given known sensitivity of the dihydropyridine ring system to hydrolytic and photolytic degradation.

3.2.P.2.2.2 Overages: No overage is proposed; the manufacturing process and analytical controls are designed to ensure label claim is met at time of release and through shelf life without the need for an intentional overage.

3.2.P.2.2.3 Physicochemical and Biological Properties: As a BCS Class I compound, in vitro dissolution is not expected to be the rate-limiting step for absorption; nonetheless, dissolution testing is included in the specification as a standard quality control and batch-to-batch consistency measure, and comparative dissolution profiling against the RLD supports the overall bioequivalence narrative documented in Module 5.

3.2.P.2.3 Manufacturing Process Development

The manufacturing process (direct compression) was selected at formulation development stage and is intended to be maintained from exhibit batch through commercial scale, subject to standard scale-up considerations (blend uniformity at scale, compression tooling change, if any). Any changes identified during scale-up or post-approval would be evaluated and managed in accordance with applicable FDA post-approval change guidance (e.g., SUPAC-IR, 21 CFR 314.70), with the reporting category (Annual Report, Changes Being Effected, or Prior Approval Supplement) determined by the nature and potential impact of the change on product quality.

3.2.P.2.4 Container Closure System

The proposed container closure system (HDPE bottle with child-resistant closure, and/or PVC/aluminium blister) is consistent with RLD presentation and standard practice for immediate-release oral solid dosage forms; suitability is supported by compendial compliance of all packaging materials.

3.2.P.2.5 Microbiological Attributes

As a non-sterile, non-aqueous immediate-release oral tablet, AMLOZA is categorized per USP <1111> as a dosage form for which routine microbial limits testing at every batch release is not typically required, provided appropriate controls are demonstrated during development. Should testing be warranted, total aerobic microbial count and total combined yeasts/molds count would be determined per USP <61>, with absence of specified objectionable organisms confirmed per USP <62>, consistent with the acceptance criteria in USP <1111> for this dosage form category.

3.2.P.2.6 Compatibility

Compatibility studies would typically be conducted during development to confirm the absence of drug-excipient interaction (e.g., via forced degradation / binary mixture studies under accelerated conditions). No experimental compatibility data are presented in this academic dossier.

3.2.P.3 Manufacture

3.2.P.3.1 Manufacturer(s)

[Commercial manufacturing site details to be finalized prior to submission.]

3.2.P.3.2 Batch Formula

The following is an example fixed quantitative (per-tablet) composition, provided for academic portfolio demonstration purposes. The 5 mg and 10 mg strengths are formulated proportionally by weight, consistent with a common-blend, proportional-formulation approach across strengths.

Ingredient (Function)	5 mg (mg/tab)	10 mg (mg/tab)	% w/w
Amlodipine Besylate USP, eq. to amlodipine (Active)	6.94	13.87	6.94%
Microcrystalline Cellulose NF (Diluent)	60.00	120.00	60.00%
Anhydrous Dibasic Calcium Phosphate USP (Diluent/Binder)	25.00	50.00	25.00%
Sodium Starch Glycolate, Type A NF (Disintegrant)	5.00	10.00	5.00%
Magnesium Stearate NF (Lubricant)	3.06	6.13	3.06%
Total tablet core weight	100.00	200.00	100.00%

Representative exhibit batch example (10 mg strength, 100,000-tablet batch — consistent with FDA's minimum exhibit batch size recommendation of 100,000 units or 10% of the intended commercial batch size, whichever is larger, for solid oral dosage forms):

Amlodipine Besylate USP	1.387 kg
Microcrystalline Cellulose NF	12.000 kg
Anhydrous Dibasic Calcium Phosphate USP	5.000 kg
Sodium Starch Glycolate, Type A NF	1.000 kg
Magnesium Stearate NF	0.613 kg
Total batch weight	20.000 kg

The 5 mg strength batch formula scales proportionally from the same core-weight ratio. Figures are rounded to the nearest 0.01 mg / 0.001 kg for tabulation; minor rounding differences (≤ 0.01%) may occur and would be reconciled in the actual approved master batch record.

3.2.P.3.3 Description of Manufacturing Process and Process Controls

1. Dispensing of drug substance and excipients per batch formula. 2. Sifting of drug substance and diluents through appropriate mesh screen. 3. Blending of sifted materials (drug substance, diluents, disintegrant) in a suitable blender for a validated time/speed. 4. Lubrication: addition of magnesium stearate, blended for a shorter, controlled time to avoid over-lubrication. 5. Compression on a rotary tablet press to target weight, hardness, and thickness. 6. Packaging into the proposed container closure system.

3.2.P.3.4 Controls of Critical Steps and Intermediates

Step	Critical control
Blending	Blend uniformity (content uniformity of drug substance across blend samples)
Lubrication	Lubrication time controlled to prevent tablet hardness/dissolution impact from over-blending with magnesium stearate
Compression	In-process tablet weight, hardness, thickness, and friability monitored at defined intervals

3.2.P.3.5 Process Validation and/or Evaluation

Process validation will be conducted in accordance with the three-stage lifecycle approach described in FDA's guidance, Process Validation: General Principles and Practices (2011): Stage 1 (Process Design), Stage 2 (Process Qualification, including Process Performance Qualification), and Stage 3 (Continued Process Verification). Process Performance Qualification (PPQ) would be conducted on an appropriate number of commercial-scale batches manufactured using the finalized commercial process, equipment, and control strategy, evaluating blend uniformity, content uniformity, dissolution, and physical tablet attributes, prior to commercial release. [PPQ protocol and data — placeholder for academic exercise; no experimental PPQ data are available or presented.]

3.2.P.4 Control of Excipients

All excipients (microcrystalline cellulose NF, dibasic calcium phosphate USP, sodium starch glycolate Type A NF, magnesium stearate NF) comply with their respective current compendial monographs. No novel excipients are used; additional testing beyond standard incoming raw material release testing against compendial specifications is not currently proposed.

3.2.P.5 Control of Drug Product

3.2.P.5.1 Specification

Description	White, round, biconvex, uncoated tablets, debossed as applicable
Identification	HPLC retention time matches reference standard; UV/IR conforms
Assay	95.0–105.0% of label claim, by HPLC
Content uniformity	Per USP <905>, Acceptance Value criteria

Dissolution	Example method (for portfolio purposes): USP Apparatus 2 (Paddle), 50 rpm; medium: 900 mL Purified Water; temperature 37.0°C ± 0.5°C; sampling at 30 minutes; acceptance criterion: Not less than 80% (Q) of the labeled amount of amlodipine dissolved in 30 minutes. The final validated method would be established during formal method development, referencing the USP Amlodipine Tablets monograph and/or FDA product-specific dissolution guidance as applicable.
Related substances	Individual degradation product ≤ 0.2%; total degradation products ≤ 1.0% (indicative; to be finalized against ICH Q3B thresholds for max daily dose)
Water content	≤ [X]% w/w by Karl Fischer (limit to be set from stability data)
Weight variation / uniformity of dosage units	Per USP <905>
Microbial limits	Per USP <1111>, <61>, <62> — routine testing at every batch not typically required for this dosage form category provided development controls are demonstrated; tested if warranted per risk assessment

3.2.P.5.2 Analytical Procedures

Compendial USP methods (Amlodipine Tablets monograph) are used for assay, dissolution, content uniformity, and related substances, using HPLC with UV detection. Dissolution testing employs the USP-specified apparatus (Apparatus I or II) and medium.

3.2.P.5.3 Validation of Analytical Procedures

Compendial methods are verified for the specific laboratory/instrumentation per USP <1225> prior to routine use. Any non-compendial method would be validated per ICH Q2(R1).

3.2.P.5.4 Batch Analyses

Release testing data from exhibit and validation batches would be presented here in a real submission, demonstrating consistent conformance to specification. [Placeholder — batch data pending actual manufacture.]

3.2.P.5.5 Characterisation of Impurities

Degradation products are characterized against known USP-listed impurities for Amlodipine Tablets; any unknown/unspecified degradation product exceeding the identification threshold would be structurally characterized per ICH Q3B(R2).

3.2.P.5.6 Justification of Specification

The proposed specification is based on the current USP monograph for Amlodipine Tablets, supplemented with dissolution and related-substance limits informed by FDA product-specific guidance for amlodipine besylate, where applicable.

3.2.P.6 Reference Standards or Materials

As described in Module 3.2.S.5; the same USP Reference Standard is used for drug product release and stability testing.

3.2.P.7 Container Closure System

HDPE bottles with child-resistant, induction-sealed closures, and/or PVC/aluminium blister strips. Both options are consistent with materials used for the RLD and other approved amlodipine besylate generic products. Container closure suitability is supported by compendial compliance and moisture vapor transmission rate (MVTR) considerations relevant to product stability.

3.2.P.8 Stability

3.2.P.8.1 Stability Summary and Conclusion

A stability program per ICH Q1A(R2) is proposed for exhibit/validation batches: long-term storage at 25°C ± 2°C / 60% RH ± 5% RH (0, 3, 6, 9, 12, 18, 24, 36 months) and accelerated storage at 40°C ± 2°C / 75% RH ± 5% RH (0, 3, 6 months). Parameters proposed for testing: description, assay, dissolution, related substances, water content. No experimental stability data are available or presented in this academic dossier; shelf life would be established based on data generated under this or an equivalent approved protocol.

3.2.P.8.2 Postapproval Stability Protocol and Commitment

Annual batch stability commitment per ICH Q1A(R2), with at least one production batch per year placed on long-term stability, tested per the approved protocol.

3.2.P.8.3 Stability Data

Actual stability data tables (assay, dissolution, and related substances trended by time point and condition) would be presented here upon generation of real batch data. Not fabricated for this academic exercise.

*Portfolio Note: Section 3.2.P.2 (Pharmaceutical Development) is the strongest demonstration of applied regulatory/formulation reasoning in this dossier — it explains *why* each formulation and process decision was made, not just what the formula is. This is the section most likely to be probed in an interview.*